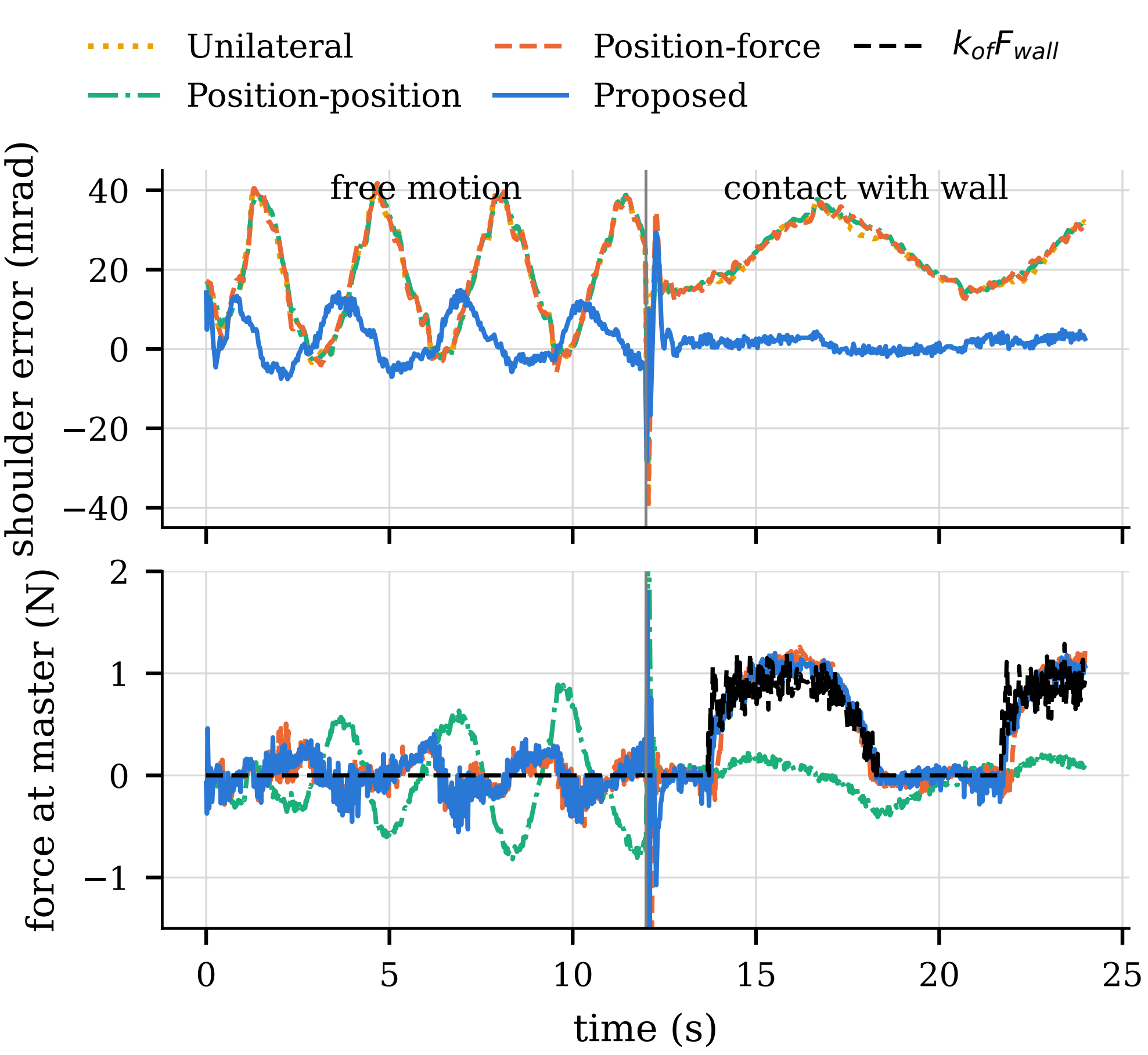

Unilateral
Position-force
$k_{of}F_{wall}$
Position-position
Proposed
free motion
contact with wall
shoulder error (mrad)
40
20
0
−20
−40
force at master (N)
2
1
0
−1
0
5
10
15
20
25
time (s)